Q:- A square matrix is invertible if and only if it is non singular matrix.

Solution:- Let A is a square matrix of order n

Suppose that A is invertible then there exists a square matrix B of order n such that

$$AB = I$$

$$\Rightarrow \det(AB) = \det(I)$$

$$\Rightarrow (\det A)(\det B) = 1 \neq 0$$

$$\Rightarrow \det A \neq 0$$

Thus A is non-singular matrix.

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Conversely, suppose that A is a non singular matrix. i.e. $|A| \neq 0$.
we have

$$A(\text{adj} A) = (\text{adj} A) A = |A| I$$

Dividing both sides by $|A|$, we get

$$A\left(\frac{\text{adj} A}{|A|}\right) = \left(\frac{\text{adj} A}{|A|}\right) A = I$$

Thus A is invertible with $A^{-1} = \frac{\text{adj} A}{|A|}$.

Hence proved.

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